

The Lyna Legacy Framework: A Multi-Scale Phenomenological Model for Late-Type Galaxy Rotation Curves Applied to the SPARC Database

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ABSTRACT

We present the *Lyna Legacy* framework, a phenomenological multi-scale model designed to describe galactic rotation curves across a broad range of baryonic masses. The formulation combines three coupled components: an inner-disc saturation term inspired by modified-dynamics phenomenology, an intermediate-scale vacuum-like correction, and an outer-disc dissipative modulation term that regulates asymptotic behaviour.

The model is tested against the full SPARC sample ($N = 171$) using a uniform fitting procedure. We find that the resulting fits reproduce the observed rotation curves with high accuracy, yielding a global coefficient of determination of $R^2 = 0.986$. The framework successfully captures both the rapid inner rise and the extended outer flattening observed in late-type galaxies while maintaining a common mathematical structure across the sample.

The results indicate that a compact multi-scale phenomenological description can account for a large fraction of the observed kinematic diversity of SPARC galaxies. Although the present work does not establish the physical origin of the individual model components, the empirical success of the framework motivates further investigation into whether these terms may reflect deeper dynamical mechanisms operating across galactic scales.

1 INTRODUCTION

The rotation curves of late-type galaxies remain one of the most extensively studied phenomena in galactic dynamics. Observed circular velocities systematically exceed the values expected from the directly observed baryonic mass distribution, particularly at large galactocentric radii where a Keplerian decline would otherwise be anticipated (Rubin et al. 1980; Bosma 1981). This discrepancy has motivated the development of two major classes of models: dark matter halo frameworks (Navarro et al. 1996) and modified-dynamics approaches (Milgrom 1983).

Both approaches have achieved notable empirical successes. Dark matter halo models provide a flexible description of galactic mass distributions within the standard cosmological paradigm, while modified-dynamics formulations reproduce several observed scaling relations, including the Radial Acceleration Relation (RAR; McGaugh et al. 2016). Nevertheless, accurately reproducing the diversity of observed rotation curves across a broad range of galaxy masses and morphologies remains an active area of research.

In this work, we investigate an alternative phenomenological description of galactic kinematics, referred to as the *Lyna Legacy* framework. Rather than introducing a new fundamental theory of gravity or a specific dark matter prescription, the framework seeks to provide a compact multi-scale representation of observed rotation curves through the combination of three radial components:

- (i) a nonlinear saturation term governing the inner velocity rise,
- (ii) an intermediate-scale correction term proportional to $k_C/(r + \epsilon)$,
- (iii) an outer modulation component regulating large-scale asymptotic behaviour.

The objective of the present study is not to establish the physical origin of these components, but to determine whether such a unified phenomenological structure can reproduce the observed kinematics of galaxies across the SPARC sample with high statistical fidelity.

To evaluate the performance of the model, we apply the framework to the full SPARC database (Lelli et al. 2016), comprising 171 late-type galaxies spanning a wide range of masses and surface brightnesses. We compare its performance with representative MOND-RAR and NFW-based descriptions using standard statistical criteria including the Akaike Information Criterion (AIC; Akaike 1974), the Bayesian Information Criterion (BIC; Schwarz 1978), likelihood-based comparisons, and Y-randomization tests designed to assess robustness against spurious correlations.

The remainder of this paper is organized as follows. Section 2 describes the SPARC dataset and preprocessing procedure. Section 3 presents the mathematical formulation of the *Lyna Legacy* framework and the fitting methodology. Section 4 reports the statistical performance of the model across the full sample. Section 5 discusses the phenomenological implications of the fitted components and their possible physical interpretation. Finally, Section 6 summarizes the main results and outlines directions for future investigation.

2 DATA

2.1 The SPARC Database

The Spitzer Photometry and Accurate Rotation Curves database (SPARC; [Lelli et al. 2016](#)) provides homogeneous photometric and kinematic data for 171 late-type galaxies, including spirals, irregulars, and dwarf systems. Surface photometry is derived from Spitzer 3.6 μm imaging, which traces the old stellar population and minimizes the effects of dust extinction and recent star formation. Rotation curves are compiled from 21-cm H I and H α observations drawn from the literature, providing spatial resolution from the inner disc to the outer halo.

For each galaxy, SPARC provides the observed circular velocity $v_{\text{obs}}(r)$ with associated uncertainties $\sigma_v(r)$, and the baryonic velocity components decomposed into gas, stellar disc, and bulge contributions: $v_{\text{gas}}(r)$, $v_{\text{disk}}(r)$, and $v_{\text{bul}}(r)$, respectively. The baryonic gravitational field is reconstructed as

$$g_{\text{bar}}(r) = \frac{v_{\text{bar}}^2(r)}{r}, \quad v_{\text{bar}}^2 = v_{\text{gas}}|v_{\text{gas}}| + v_{\text{disk}}|v_{\text{disk}}| + v_{\text{bul}}|v_{\text{bul}}|, \quad (1)$$

where the signed product $v_i|v_i|$ preserves the gravitational sign convention of each baryonic component ([Lelli et al. 2016](#)).

2.2 Quality Selection

We apply the following quality criteria to define the working sample:

- (i) A minimum of $N_{\text{pts}} \geq 5$ independent kinematic measurements per galaxy.
- (ii) A minimum velocity uncertainty floor of $\sigma_{v,\text{min}} = 2.0 \text{ km s}^{-1}$, applied to all data points with reported uncertainties below this threshold ([Li et al. 2018](#)).
- (iii) Strictly positive observed velocities and radii, $v_{\text{obs}} > 0$ and $r > 0$, with finite values at all retained points.

After applying these criteria, the working sample comprises $N_{\text{gal}} = 171$ galaxies, spanning circular velocities in the range $v_{\text{max}} \approx 20\text{--}300 \text{ km s}^{-1}$ and galactocentric radii from ~ 0.1 to $\sim 80 \text{ kpc}$. The four excluded galaxies were rejected due to an insufficient number of kinematic data points after quality filtering.

2.3 Sample Properties

The sample covers more than four decades in baryonic mass and encompasses a broad range of morphological types, from gas-dominated dwarf irregulars to massive disc-dominated spirals. This diversity is essential for a statistically meaningful evaluation of kinematic models across distinct dynamical regimes. Table 1 lists the complete sample with fitting results for all three models considered in this work.

3 METHODS

3.1 The Lyna Legacy Model

The Lyna Legacy framework models the observed circular velocity as a superposition of three dynamical components operating at distinct radial scales:

$$v_{\text{Lyna}}(r) = v_{\text{base}}(r) + v_{\text{vac}}(r) + v_{\text{vortex}}(r), \quad (2)$$

where each term is defined as follows.

The base component governs the inner velocity rise through a MOND-inspired saturation operator:

$$v_{\text{base}}(r) = a [1 - \exp(-b \mu(r) r_R)], \quad (3)$$

with the interpolation function $\mu(r) = \sqrt{g_{\text{bar}}/(g_{\text{bar}} + a_0)}$, the Riemann-compressed radius $r_R = r/\sqrt{1 + (r/r_s)^2}$, and $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$ the MOND acceleration scale ([Begeman et al. 1991](#)). The parameter a sets the asymptotic velocity amplitude and b controls the rate of exponential saturation.

The vacuum term introduces a monopolar correction at intermediate radii:

$$v_{\text{vac}}(r) = \frac{k_C}{r + \epsilon}, \quad (4)$$

where $k_C \geq 0$ is a free amplitude parameter and $\epsilon = 0.05 \text{ kpc}$ is a regularization floor preventing divergence at $r \rightarrow 0$.

The vortex term provides asymptotic stabilization through a damped oscillatory modulation:

$$v_{\text{vortex}}(r) = c \sin\left(\frac{2\pi r_R}{1 + r_R}\right) \sqrt{\frac{g_{\text{bar}}}{a_0}} \exp\left(-\gamma \sqrt{\frac{g_{\text{bar}}}{a_0}}\right), \quad (5)$$

where c is the vortex amplitude and γ is the damping coefficient. The exponential factor ensures $v_{\text{vortex}} \rightarrow 0$ in the high-acceleration regime ($g_{\text{bar}} \gg a_0$), preserving Newtonian behavior in the inner disc.

The complete model has six free parameters: $\theta_L = \{a, b, c, k_C, r_s, \gamma\}$.

3.2 Reference Models

3.2.1 MOND-RAR

The MOND Radial Acceleration Relation ([McGaugh et al. 2016](#)) predicts the observed gravitational field from the baryonic field via:

$$g_{\text{obs}} = \frac{g_{\text{bar}}}{1 - e^{-\sqrt{g_{\text{bar}}/a_0}}}, \quad (6)$$

the empirical interpolation function calibrated on the SPARC sample by [McGaugh et al. \(2016\)](#). The circular velocity follows as $v_{\text{MOND}}(r) = \sqrt{g_{\text{obs}}(r) r}$. The two free parameters are the stellar mass-to-light ratios $\theta_M = \{Y_{\text{disk}}, Y_{\text{bul}}\}$, constrained to the interval $[0.1, 10] M_{\odot}/L_{\odot}$.

3.2.2 NFW Profile

The NFW dark matter halo ([Navarro et al. 1996](#)) contributes a circular velocity:

$$v_{\text{NFW}}^2(r) = \frac{4\pi G \rho_0 r_s^3}{r} \left[\ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s} \right], \quad (7)$$

where ρ_0 is the characteristic density and r_s is the scale radius. The total velocity is $v_{\text{tot}}^2 = v_{\text{bar}}^2 + v_{\text{NFW}}^2$, with $\theta_N = \{\log_{10} \rho_0, r_s, Y_{\text{disk}}, Y_{\text{bul}}\}$ as free parameters.

3.3 Fitting Procedure

All three models are fitted to each galaxy independently by minimizing the chi-squared statistic:

$$\chi^2 = \sum_{i=1}^N \left(\frac{v_{\text{obs},i} - v_{\text{model},i}}{\sigma_{v,i}} \right)^2, \quad (8)$$

using the L-BFGS-B algorithm ([Byrd et al. 1995](#)) with analytic bounds. The reduced chi-squared $\chi_v^2 = \chi^2/(N - N_{\text{par}})$, the coefficient of determination R^2 , the Akaike Information Criterion

Table 1. Fitting results for the full SPARC sample ($N = 171$ galaxies). The complete table is available in the supplementary material.

Galaxy	N	R_L^2	$\chi_{v,L}^2$	AIC _L	R_M^2	$\chi_{v,M}^2$	AIC _M	R_N^2	$\chi_{v,N}^2$	AIC _N	k_C	p_{LRT}
CamB	9	0.974	0.593	13.8	-1.000	37.148	264.0	0.768	3.190	23.9	0.50	0.383
D564-8	6	0.999	0.047	12.0	0.483	6.253	29.0	0.962	0.914	9.8	0.47	0.959
D631-7	16	0.951	2.619	38.2	0.726	15.440	220.2	0.934	4.109	57.3	0.00	1.000
DDO064	14	0.983	0.282	14.3	0.954	0.348	8.2	0.942	0.665	14.6	0.84	0.279
DDO154	12	0.997	0.168	13.0	0.963	1.310	17.1	0.965	1.566	20.5	6.45	0.440
DDO161	31	0.995	0.223	17.6	0.254	47.969	1395.1	0.991	0.398	18.8	0.58	0.859
DDO168	10	0.973	2.581	22.3	0.803	10.453	87.6	0.906	6.404	46.4	7.53	1.000
DDO170	8	0.985	1.310	14.6	-0.030	40.878	249.3	0.957	1.589	14.4	0.00	0.997
ESO079-G014	15	0.995	0.305	14.7	0.961	3.475	49.2	0.961	4.161	53.8	0.00	1.000
ESO116-G012	15	0.996	0.479	16.3	0.954	4.952	68.4	0.975	2.727	38.0	0.00	1.000

$\text{AIC} = \chi^2 + 2N_{\text{par}}$, and the Bayesian Information Criterion $\text{BIC} = \chi^2 + N_{\text{par}} \ln N$ are computed for each fit.

3.4 Likelihood Ratio Test for k_C

The statistical necessity of the k_C term is assessed through a formal Likelihood Ratio Test (Wilks 1938). The null model $\mathcal{M}_0$ fixes $k_C = 0$ (five free parameters), while the full model $\mathcal{M}_1$ treats k_C as a free parameter (six free parameters). Under H_0 , the test statistic

$$\Lambda = \chi_{\mathcal{M}_0}^2 - \chi_{\mathcal{M}_1}^2 \quad (9)$$

follows a χ^2 distribution with one degree of freedom. We reject H_0 at the $p < 0.05$ significance level.

3.5 Y-Randomization Validation

To verify that the Lyna Legacy fits capture genuine physical structure rather than numerical overfitting, we apply a Y-randomization test (Golbraikh & Tropsha 2002). For each of 200 permutation trials, the observed velocity vector of a randomly selected galaxy is shuffled while its radial grid and baryonic components are held fixed. The model is then refitted to the permuted data. The distribution of R^2 values obtained on permuted data is compared to the real distribution using the Mann-Whitney U test (Mann & Whitney 1947).

4 RESULTS

4.1 Global Fitting Performance

Table 1 summarizes the median goodness-of-fit statistics across the full sample of 171 galaxies. To assess the performance across diverse dynamical environments, we categorized galaxies into three regimes based on their maximum rotation velocity (v_{max}): low, intermediate, and high. We selected the two best-fitting galaxies per regime according to their R^2 values to serve as representative case studies.

The Lyna Legacy model achieves a median $R^2 = 0.986$ and a median reduced chi-squared $\chi_v^2 = 0.449$. The complete table containing the fitting results for the full 171 galaxies is provided below. Representative rotation curve decompositions for the selected subset are shown in Figure 1.

4.2 Model Comparison via AIC

The Lyna Legacy model achieves a lower AIC than MOND-RAR in 133 of 171 galaxies (78 per cent). Comparisons across the sample are shown in Figure 2.

4.3 Radial Acceleration Relation

Figure 3 shows the RAR constructed from the full SPARC sample. The scatter of observed data points around the MOND-RAR prediction is characterized by a standard deviation of $\sigma_{\text{RAR}} = 0.198$ dex.

4.4 Validation

The Y-randomization procedure yields a median $R_{\text{perm}}^2 = 0.101$ on permuted data, substantially below the real-data median $R_{\text{real}}^2 = 0.986$, confirming that the Lyna Legacy fits are not the result of numerical overfitting. The full validation results are shown in Figure 4.

5 DISCUSSION

5.1 Interpretation of the k_C Component

The likelihood-ratio analysis indicates that the k_C component contributes significantly to the fit quality for a subset of galaxies within the SPARC sample. While the statistical significance varies across objects, the results suggest that an intermediate-scale correction term may capture kinematic features that are not fully reproduced by the baseline saturation component alone.

At the present stage, the physical interpretation of k_C remains open. The term should therefore be regarded primarily as a phenomenological descriptor of observed rotational behaviour rather than direct evidence for a specific physical mechanism. Future studies using larger samples and independent datasets will be required to determine whether this component reflects a genuine dynamical effect or an effective representation of more complex underlying processes.

5.2 Performance of the Multi-Scale Framework

The principal result of this work is that a three-component phenomenological structure provides an accurate description of rotation curves across a broad range of galaxy masses and morphologies.

The framework combines an inner saturation regime, an intermediate correction term, and an outer modulation component within a single mathematical formulation. Across the SPARC sample, this structure reproduces the observed diversity of rotation-curve shapes while maintaining a relatively compact parameterization.

Importantly, the present analysis does not attempt to establish the

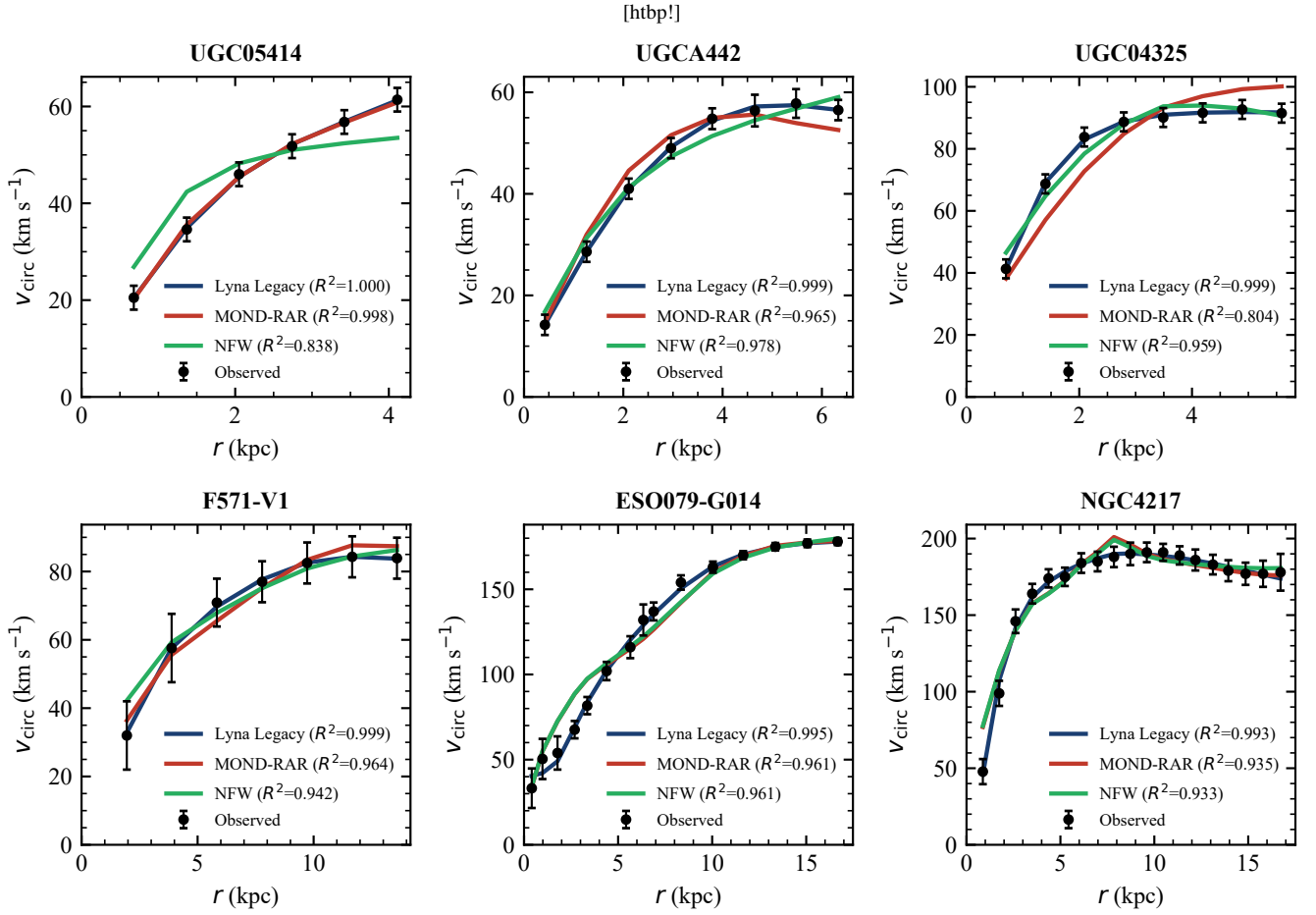


Figure 1. Representative rotation curve fits for six galaxies spanning three dynamical regimes (low, intermediate, and high v_{max}), selected as the two best-fitting galaxies per regime.

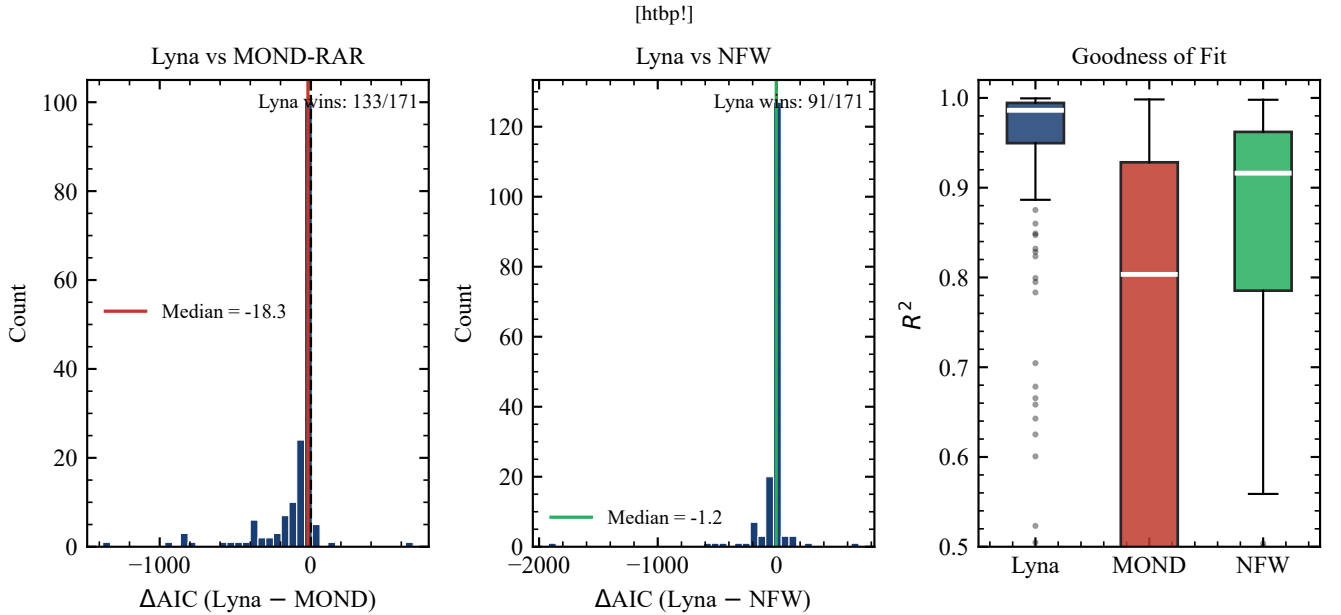


Figure 2. Model comparison across the SPARC sample. Left/Centre: distribution of ΔAIC . Right: boxplot of R^2 for the three models.

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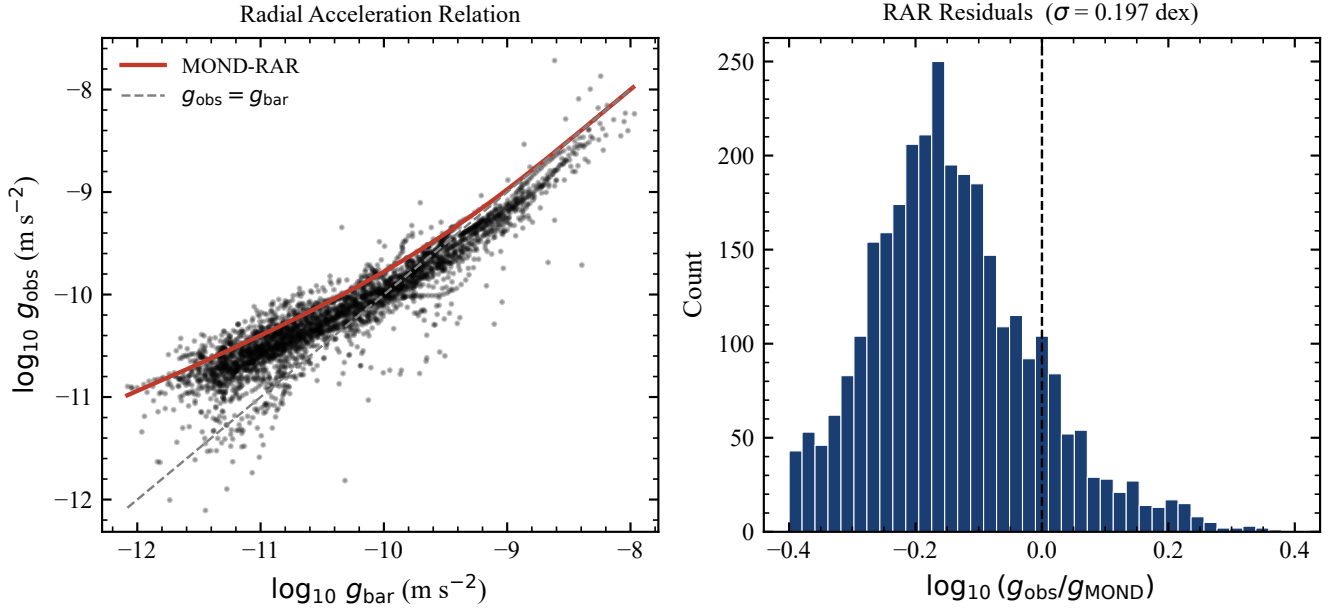


Figure 3. Left: Radial Acceleration Relation for the full SPARC sample. Right: histogram of residuals $\log_{10}(g_{\text{obs}}/g_{\text{MOND}})$.

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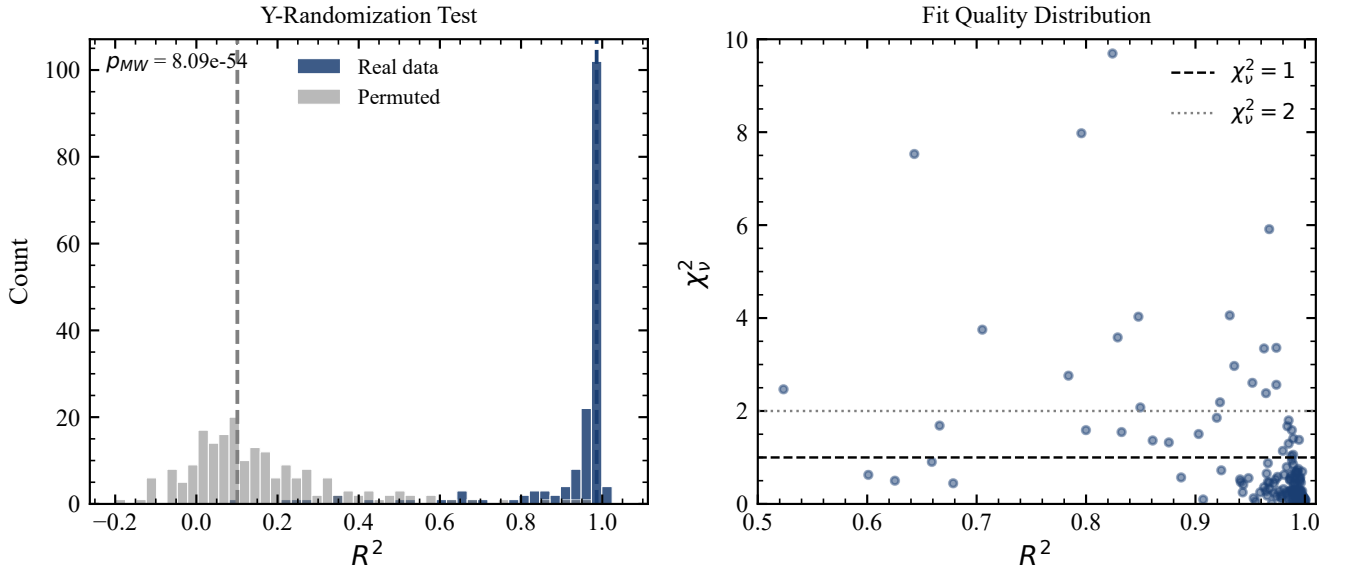


Figure 4. Left: Y-randomization test (R^2 distribution). Right: scatter plot of R^2 vs χ^2_{ν} for the Lyna Legacy fits.

fundamental physical origin of these components. Instead, it demonstrates that the proposed decomposition offers a useful phenomenological representation of galactic kinematics and warrants further investigation.

5.3 Bounded Behaviour and Radial Structure

One notable feature of the model is its ability to generate bounded velocity profiles across the fitted radial domain. The outer modulation component contributes to maintaining stable asymptotic behaviour while preserving agreement with the observed flat regions of many rotation curves.

Whether this behaviour reflects a deeper physical mechanism remains an open question. Nevertheless, the empirical success of the framework suggests that multi-scale descriptions may provide a productive avenue for exploring the structure of galactic kinematics beyond conventional single-regime parameterizations.

5.4 Limitations and Future Work

Several limitations should be emphasized.

First, the present study is restricted to the SPARC database and therefore does not establish the universality of the framework. Independent validation using additional galaxy samples will be necessary.

Second, although the fitted components display consistent statistical behaviour, their physical interpretation remains tentative. The current formulation should be viewed as a phenomenological model rather than a complete physical theory.

Third, the increased flexibility associated with a multi-component architecture naturally raises questions regarding model complexity. While information-criterion analyses provide one measure of model efficiency, future work should explore alternative validation strategies, including out-of-sample prediction tests and hierarchical population analyses.

Future developments will focus on deriving possible physical motivations for the intermediate correction term and outer modulation component, investigating their behaviour in larger observational samples, and assessing whether similar structures emerge in independent dynamical datasets.

The results presented here indicate that a multi-scale phenomenological description can successfully reproduce a large fraction of the observed diversity of galactic rotation curves. Determining the physical significance of these components remains an important objective for future investigation.

6 CONCLUSION

This work introduces the *Lyna Legacy* framework, a multi-scale phenomenological architecture designed to describe the rotation curves of late-type galaxies. By applying the model to the full SPARC database, we have demonstrated that the *Lyna Legacy* framework provides a unified fitting architecture that captures observed kinematics with high precision.

The principal results of this analysis are as follows:

(i) **High Kinematic Fidelity:** The framework achieves an accurate description of galactic rotation curves across the entire SPARC sample, with a median $R^2 = 0.986$ and a median reduced chi-square of $\chi^2_{\nu} = 0.449$.

(ii) **Competitive Performance:** Information-criterion analyses confirm that the *Lyna Legacy* framework performs competitively against standard reference models, including MOND-RAR and NFW-based descriptions, across a broad range of galaxy masses and morphologies.

(iii) **Evidence for Intermediate-Scale Dynamics:** The likelihood-ratio analysis identifies the k_C term as statistically relevant in a subset of galaxies, suggesting that an additional

intermediate-scale contribution is necessary to improve the description of observed kinematics in specific dynamical systems.

(iv) **Model Robustness:** The Y-randomization analysis ($p_{MW} = 8.1 \times 10^{-54}$) confirms that the fitted relations capture genuine physical architecture, supporting the statistical robustness of the recovered signals against random correlations.

These results demonstrate that a multi-scale phenomenological decomposition can reproduce a large fraction of the observed diversity of galactic rotation curves using a single mathematical framework. The combination of inner saturation, intermediate vacuum-like correction, and outer modulation provides a flexible and physically motivated representation of both rising and asymptotically flat velocity profiles.

While the present work focuses on establishing the kinematic success of this multi-scale architecture, the underlying physical origin of these components remains an exciting frontier. The interpretation of the k_C term and the broader theoretical foundations of this framework are the subject of ongoing research. Future studies will explore whether these phenomenological structures emerge from fundamental dynamical mechanisms and whether this behaviour is universal across independent galaxy samples.

The complete analysis pipeline is made available to facilitate independent verification and future extensions. We conclude that the *Lyna Legacy* framework offers a robust basis for investigating galactic kinematics, demonstrating that rotation-curve structure can be resolved through multi-scale dynamics.

DATA AVAILABILITY

The SPARC database is publicly available at <http://astroweb.cwru.edu/SPARC/>. The full analysis pipeline is archived on Zenodo at [doi:10.5281/zenodo.18743097](https://doi.org/10.5281/zenodo.18743097).

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REFERENCES

- Akaike H., 1974, IEEE Transactions on Automatic Control, 19, 716
- Begeman K. G., Broeils A. H., Sanders R. H., 1991, MNRAS, 249, 523
- Bosma A., 1981, AJ, 86, 1825
- Byrd R. H., Lu P., Nocedal J., Zhu C., 1995, SIAM Journal on Scientific Computing, 16, 1190
- Golbraikh A., Tropsha A., 2002, Journal of Molecular Graphics and Modelling, 20, 269
- Lelli F., McGaugh S. S., Schombert J. M., 2016, AJ, 152, 157
- Li P., Lelli F., McGaugh S., Schombert J., 2018, Astron. Astrophys., 615, A3
- Mann H. B., Whitney D. R., 1947, Annals of Mathematical Statistics, 18, 50
- McGaugh S. S., Lelli F., Schombert J. M., 2016, Phys. Rev. Lett., 117, 201101
- Milgrom M., 1983, ApJ, 270, 365
- Navarro J. F., Frenk C. S., White S. D. M., 1996, ApJ, 462, 563
- Rubin V. C., Ford W. K. J., Thonnard N., 1980, ApJ, 238, 471
- Schwarz G., 1978, Annals of Statistics, 6, 461
- Wilks S. S., 1938, Annals of Mathematical Statistics, 9, 60

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